

## Scalability & Future Plan

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 10 July 2026                                                 |
| Team ID       |                                                              |
| Project Name  | OptiCrop : Smart Agricultural Production Optimization Engine |
| Maximum Marks | 1 Mark                                                       |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                                                              | Impact                                                          | Priority to Address (High / Medium / Low) |
|------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|
| 1    | Application currently runs on a local Flask server and is not deployed for public access.                               | Limits accessibility to local users only.                       | High                                      |
| 2    | Crop recommendation is based only on soil nutrients and environmental parameters without real-time weather information. | Recommendations may not fully reflect current field conditions. | High                                      |
| 3    | The system does not provide fertilizer recommendations or disease detection features.                                   | Farmers require additional guidance beyond crop selection.      | Medium                                    |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                  | Proposed Upgrade / Solution                                                                                                              |
|------|--------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a single local user.                                  | Deploy on cloud platforms such as Render, Railway, or Azure to support multiple concurrent users.                                        |
| 2    | Data Storage       | Uses a static CSV dataset for training.                        | Integrate a relational database such as PostgreSQL or MySQL to store user inputs, prediction history, and updated agricultural datasets. |
| 3    | Performance        | Predictions are generated using a locally stored Pickle model. | Optimize the model, implement caching, and use production-grade web servers for faster response under higher traffic.                    |
| 4    | Security           | Basic client-side and server-side input validation             | Add user authentication, HTTPS, secure API endpoints, and role-based access control for enhanced security.                               |

## Future Roadmap:

| Phase          | Planned Feature / Enhancement                                                                     | Target Timeline | Expected Impact                                                              |
|----------------|---------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------|
| <b>Phase-1</b> | Deploy the application on a cloud platform and improve the user interface                         | 1-2 Months      | Enable public access and improve usability.                                  |
| <b>Phase-2</b> | Integrate real-time weather APIs and provide weather-aware crop recommendations.                  | 3-5 Months      | Increase recommendation accuracy by incorporating live environmental data.   |
| <b>Phase-3</b> | Add fertilizer recommendation, soil health analysis, and crop image support.                      | 3-5 Months      | Offer comprehensive agricultural decision support for farmers.               |
| <b>Phase-4</b> | Develop Android and iOS mobile applications with multilingual support and IoT sensor integration. | 6 Months        | Improve accessibility, scalability, and support for smart farming practices. |

## Conclusion:

The current OptiCrop system provides an accurate machine learning-based crop recommendation solution using soil and environmental parameters. Future enhancements will focus on cloud deployment, real-time weather integration, fertilizer recommendations, mobile accessibility, and smart farming technologies to improve scalability, usability, and agricultural productivity.